

	Course Name:	Electrical Network Analysis	Course Code:	EE-2004
	Program:	BS (EE)	Semester:	Fall 2023
	Duration:	180 Minutes	Total Marks:	100
	Paper Date:	21-Dec-2023	Weight:	50%
	Section:	All	Page(s):	3
	Exam Type:	Final	CLO(s)	1,2,3,4,5

Student Name: Haseeb ur Rehman Roll No. 22L-6234 Section: 3A

Instruction/Notes:

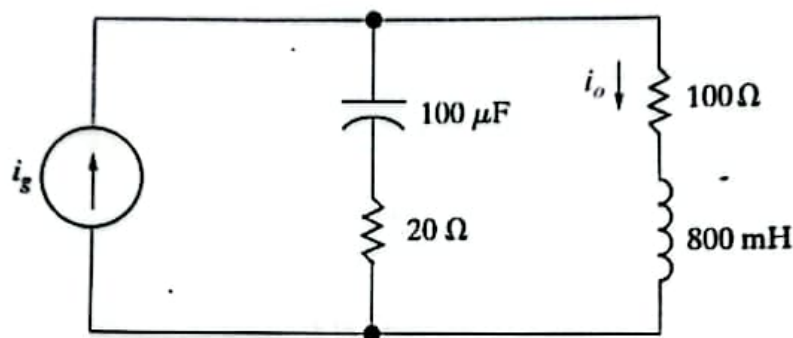
- Attempt all the questions and solve all parts of a question together
- Carefully understand the question before attempting
- Show all calculations, direct answers are not acceptable.

Question # 1

CLO 1

[20 marks]

Apply phasor-domain analysis to determine the steady state expressions for i_o if $i_g = 60 \cos(250t)$ mA.



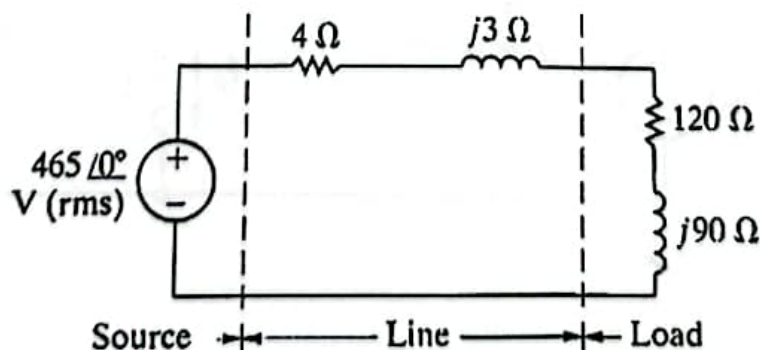
Question # 2

CLO 2

[20marks]

For the circuit shown below, calculate:

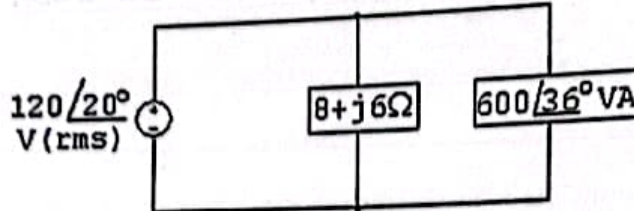
- The rms current (I_{rms}) flowing through the circuit.
- The complex power delivered by the source.
- The average power and the reactive power dissipated across the line.
- The average power and the reactive power dissipated across the load. Also, construct its power triangle.
- The value of capacitor, to be included in parallel with the load, to correct load's power factor to unity if the source frequency is 60 Hz.



CLO 3

Question #3

In a balanced three-phase system, the source is a balanced-Y with an abc phase sequence and a voltage $V_{an} = 120 \angle 20^\circ \text{ V}_{\text{rms}}$. The source feeds two loads, both of which are Y-connected. The impedance of load 1 is $8 + j6 \Omega/\phi$. The complex power for the a-phase of load 2 is $600 \angle 36^\circ \text{ VA}$. The single-phase equivalent circuit for the a-phase is given below.



Analyze the circuit to calculate the:

- line voltages at the source end;
- line current, I_{aA} .
- total complex power supplied by the source and construct its power triangle.
- combined impedance of the loads

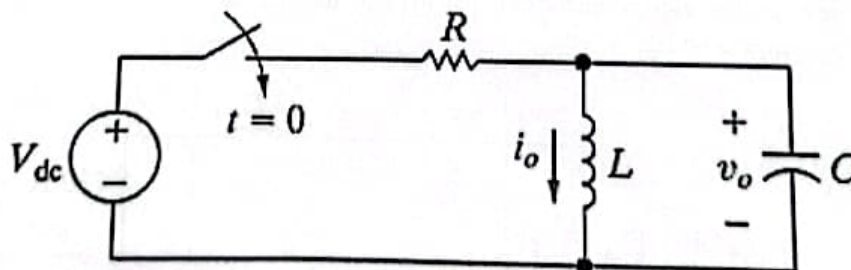
Question #4

CLO 4

[20 marks]

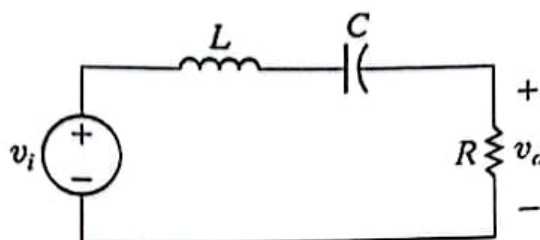
The switch in the following circuit has been open for a long time. At $t = 0$, the switch closes. Use $R = 2 \Omega$, $L = 0.8 \text{ H}$, $C = 0.25 \text{ F}$, and $V_{dc} = 5.0 \text{ V}$.

- Transform the given circuit in s-domain, for $t \geq 0$
- Derive the expression of V_o in s-domain.
- Find the time-domain expression of $v_o(t)$.
- Use the s-domain expression derived above to predict the initial- and final-values of $v_o(t)$.



Consider the following bandpass filter circuit. The filter has a center frequency, f_0 of 12.0 kHz and a quality factor, Q of 6.0. Using a $0.1 \mu\text{F}$ capacitor, analyze the circuit to find the:

- Filter's bandwidth, β .
- Values of R and L .
- Filter's upper and lower cut-off frequency, ω_{c1} and ω_{c2} .
- Simplified expression of the filter's transfer function, $H(s) = V_o(s) / V_i(s)$.
- Plot the frequency response of the filter circuit.



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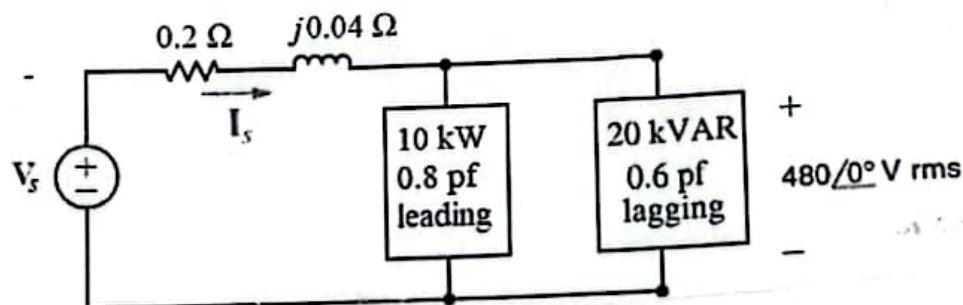
	Course Name:	Electrical Network Analysis	Course Code:	EE-2004
	Program:	BS (EE)	Semester:	Fall 2023
	Duration:	60 Minutes	Total Marks:	40
	Paper Date:	11/11/2023	Weight	15%
	Section:	All	Page(s):	1
	Exam Type:	Midterm -2	CLO	2 & 3

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- Instruction/Notes:
- Attempt all the questions and solve all parts of a question together
 - Carefully understand the question before attempting
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Question # 1 **CLO 2** **[8 + 4 + 4 + 4 = 20 marks]**

The two loads in the circuit shown below can be described as follows:
Load 1 absorbs an average power of 10.0 kW at a leading power factor of 0.8 and Load 2 absorbs 20.0 kVAR at a lagging power factor of 0.6. A voltage of $480\angle 0^\circ$ V_{rms} appears across the loads as shown.



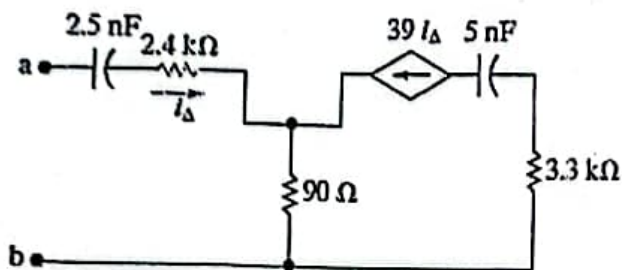
- Construct the power triangle of each load and determine the power factor of the loads in parallel. Also calculate the impedance of the combined load.
- Compute the magnitude of the current, I_s and the average power loss in the transmission line.
- Calculate the value of source voltage, V_s and the power factor of the source
- Given that the frequency of the source is 60 Hz, compute the value of the capacitor that would correct the power factor to 1.0 if placed in parallel with the loads.

Question # 2 **CLO 3** **[5 + 5 + 5 + 5 = 20 marks]**

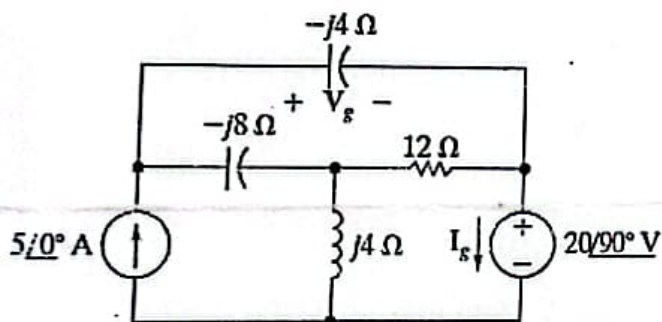
The phase voltage at the terminals of the a-phase of a balanced three-phase Y-connected load is $1600\angle 0^\circ$ V_{rms}. The load has an impedance of $12 + j9 \Omega/\phi$ and is fed from a line having an impedance of $0.1 + j1 \Omega/\phi$. The Y-connected source at the sending end of the line, has a positive phase sequence and an internal impedance of $0.02 + j0.2 \Omega/\phi$. Use the a-phase voltage at the load as the reference.

- Draw three phase and single phase diagrams of the circuit.
- Use circuit analysis techniques to compute line currents I_{aA} , I_{bB} and I_{cC} .
- Calculate line voltages at the source terminals V_{ab} , V_{bc} , and V_{ca} .
- Compute single phase and three phase active and reactive power delivered to the load.

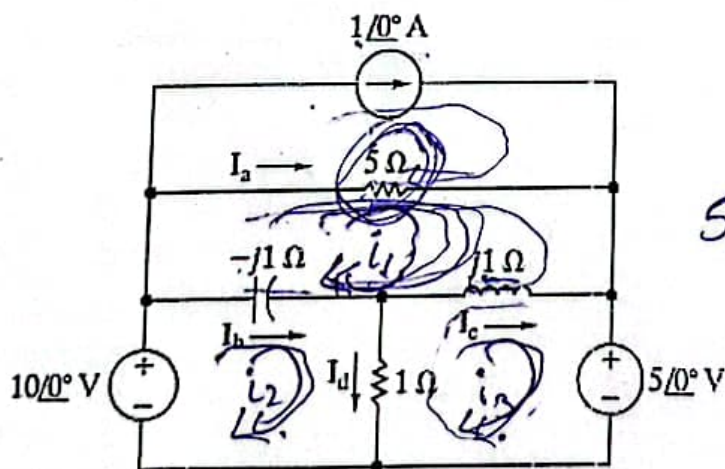
1. Find the Thévenin impedance seen looking into the terminals a,b of the circuit in following figure if the frequency of operation is $25/\pi$ kHz



2.
3. Use the node-voltage method to find the phasor voltage V_g in the circuit shown in following figure



4. Use the mesh-current method to find the branch currents in the circuit shown in following figure



$$5(i_1 - 1) + j(1)(i_1 - i_3) - j(i_1 - i_2)$$